

CSCI3230 (ESTR3108)

Fundamentals of Artificial Intelligence

Lecture 4. Support Vector Machine

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Outline

Part 1. Linear classifier

Part 2. Support Vector Machine (SVM)

Part 3. Kernel SVM

Part 4. Soft-margin SVM

Part 5. Summary



Part 1. Linear classifier

Concepts of classification

What is classification?

Concepts of classification

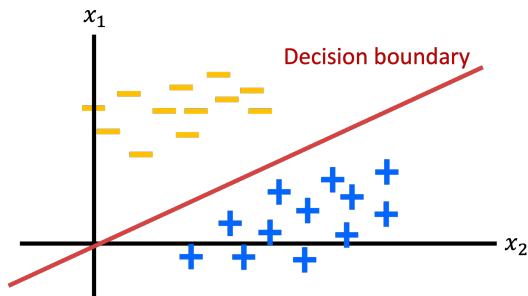
What is classification?

- **Classification** is a process related to categorization, in which ideas and objects are recognized, differentiated and understood. E.g., assigning a given email to the “spam” or “non-spam” class.
- **Goal:** Given a set of samples, with different categories, learn a classifier. Then, the classifier can automatically recognize new samples within the set of given categories.

Concepts of linear classifier

What is linear classifier?

- Linear classifier is a kind of linear models, which classifies data based on a linear combination of input features.
- Linear classifier uses a **decision boundary**, which is a hyperplane that partitions the input data into two sets, one for each class.

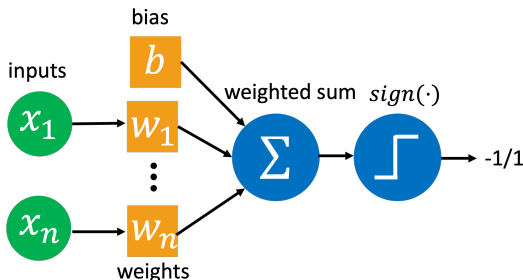


An easy binary linear classifier example

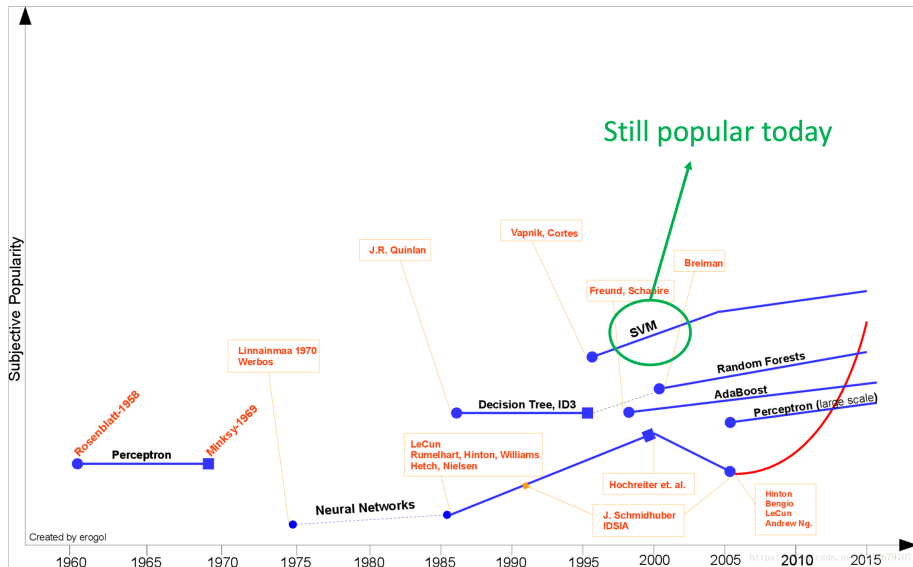
- The equation $h(x) = \mathbf{w}^T \mathbf{x} + b$ defines a hyperplane.
- One side corresponds to positive sample.
- One side corresponds to negative sample.
- Threshold the linear function $\text{sign}()$.

The prediction is positive (e.g., category “1”), if $\mathbf{w}\mathbf{x} + b > 0$

The prediction is negative (e.g., category “-1”), if $\mathbf{w}\mathbf{x} + b < 0$



History of classifier



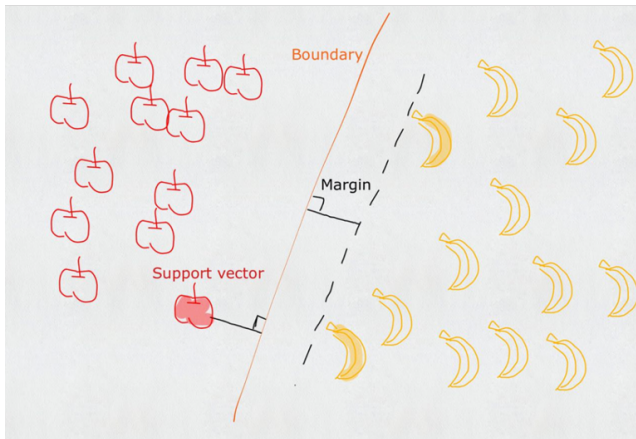
<https://erogol.com/brief-history-machine-learning/>



Part 2. Support Vector Machine (SVM)

What is Support Vector Machine (SVM)?

- One of the most theoretically well motivated and practically most effective linear classifier in machine learning.



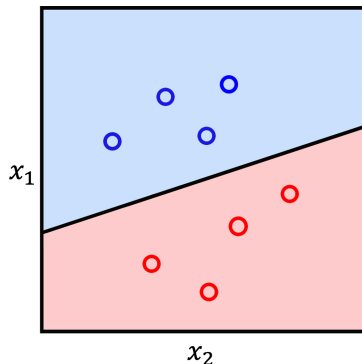
The objective of SVM

- To find a separating hyperplane as the **decision boundary**.
- Assume that we have a set of two-dimensional data:

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}$$

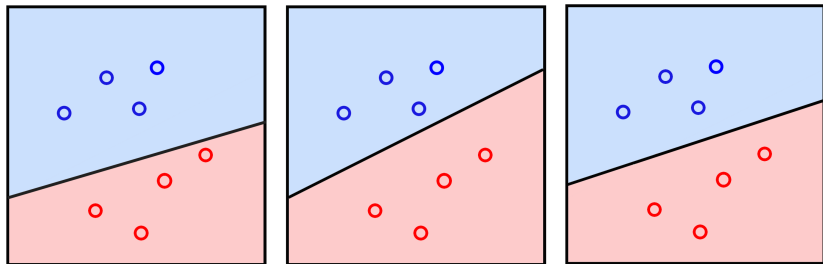
The objective of SVM

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- Assume that we have a set of two-dimensional data:
 $\mathcal{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}$
- Let's set a hyperplane: $\mathbf{w}^T \mathbf{x} + b = 0$.



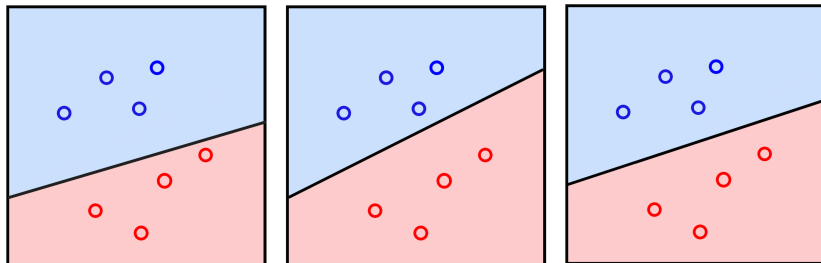
The objective of SVM

- Samples are linearly separable, which hyperplane is the “best”?



The objective of SVM

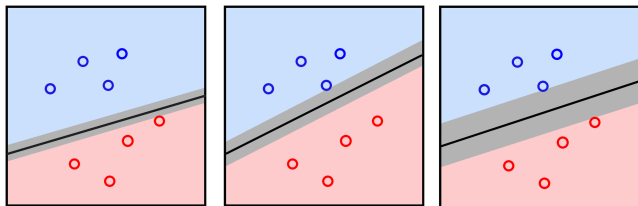
- Samples are linearly separable, which hyperplane is the “best”?



- The rightmost hyperplane seems to be the best (why?)

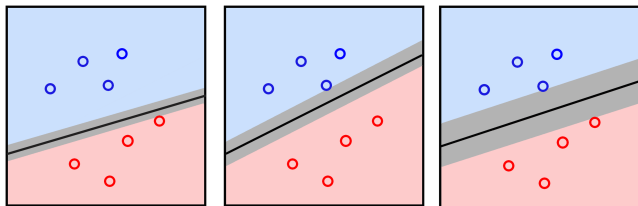
The objective of SVM

- The decision boundary has a **larger distance** to the closest point.
- The larger distance means more **noise tolerance** and therefore the classifier becomes more robust.



The objective of SVM

- Robust separating hyperplane has “**fat**” margin (i.e., far from both sides of examples).
- So, the goal is to find the **fattest** separating hyperplane.



The objective of SVM

- Assume a hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$ that can correctly classify the samples.
- Given an example point $\mathbf{x}_i$, the distance between $\mathbf{x}_i$ and the hyperplane¹ is:

$$\gamma = \frac{|\mathbf{w}^T \mathbf{x}_i + b|}{\|\mathbf{w}\|}$$

- where $\mathbf{x}_i$ and $\mathbf{w}$ are both d-dimensional vectors.

¹https://en.wikipedia.org/wiki/Distance_from_a_point_to_a_line

The objective of SVM

- To use this hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$ to classify the samples. We have:

$$y_i = \begin{cases} 1 & \text{if } \mathbf{w}^T \mathbf{x}_i + b > 0 \\ -1 & \text{if } \mathbf{w}^T \mathbf{x}_i + b < 0 \end{cases}$$

The objective of SVM

- To use this hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$ to classify the samples. We have:

$$y_i = \begin{cases} 1 & \text{if } \mathbf{w}^T \mathbf{x}_i + b > 0 \\ -1 & \text{if } \mathbf{w}^T \mathbf{x}_i + b < 0 \end{cases}$$

- Then, we have $y_i (\mathbf{w}^T \mathbf{x}_i + b) = |\mathbf{w}^T \mathbf{x}_i + b|$.
- Therefore we can transform $\gamma = \frac{|\mathbf{w}^T \mathbf{x}_i + b|}{\|\mathbf{w}\|}$ to $\gamma = \frac{y_i (\mathbf{w}^T \mathbf{x}_i + b)}{\|\mathbf{w}\|}$

The objective of SVM

- As for $(\mathbf{w}', b')$ we can transform: $\varsigma \mathbf{w} \mapsto \mathbf{w}'$ and $\varsigma b \mapsto b'$.
So that, given an arbitrary sample, we can rewrite:

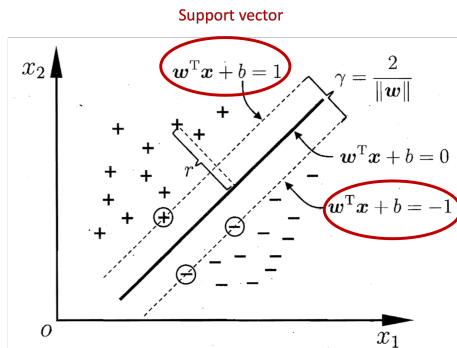
$$y_i = \begin{cases} 1 & \text{if } \mathbf{w}^T \mathbf{x}_i + b > 0 \\ -1 & \text{if } \mathbf{w}^T \mathbf{x}_i + b < 0 \end{cases}$$

$\Downarrow$

$$y_i = \begin{cases} 1 & \text{if } \mathbf{w}^T \mathbf{x}_i + b \geq 1 \\ -1 & \text{if } \mathbf{w}^T \mathbf{x}_i + b \leq -1 \end{cases}$$

The objective of SVM

- For $y_i = \begin{cases} 1 & \text{if } w^T x_i + b \geq 1 \\ -1 & \text{if } w^T x_i + b \leq -1 \end{cases}$, we can get the equal sign at the sample which is closest to the hyperplane.
- We call them **support vector**.
- At the support vector (x_s, y_s) , we have $y_s (w^T x_s + b) = 1$.

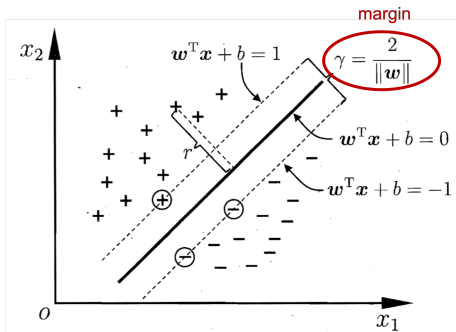


The objective of SVM

- We summarize the distances of **positive** support vector and **negative** support vector to the hyperplane, both are equal to:

$$\frac{y_s (w^T x_s + b)}{\|w\|} \Rightarrow \frac{1}{\|w\|}$$

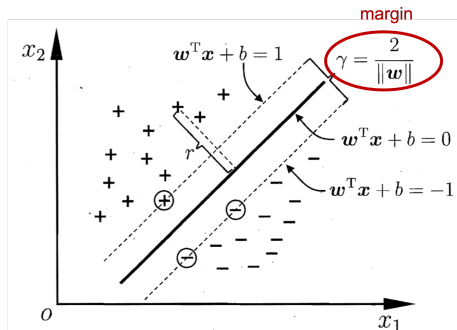
- The summarized distance is called the **margin**: $\gamma = \frac{2}{\|w\|}$.



The objective of SVM

- As we want to find the fattest separating hyperplane, our goal is to maximize the margin:

$$\begin{aligned} \max_{w,b} \quad & \frac{2}{\|w\|} \\ \text{s.t. } y_i = \quad & \begin{cases} 1 & \text{if } w^T x_i + b \geq 1 \\ -1 & \text{if } w^T x_i + b \leq -1 \end{cases} \end{aligned}$$



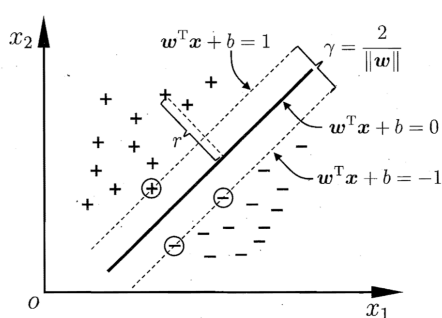
The objective of SVM

- Then, we transform the objective

$$\max \frac{2}{\|w\|} \Rightarrow \min \frac{1}{2} \|w\| \Rightarrow \min \frac{1}{2} \|w\|^2$$

- Therefore, the overall optimization objective is:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad y_i (w^T x_i + b) \geq 1$$



The objective of SVM

- Rewrite the optimization objective:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

$\Downarrow$

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{s.t.} \quad g_i(\mathbf{w}, b) = 1 - y_i (\mathbf{w}^T \mathbf{x}_i + b) \leq 0, i = 1, 2, \dots, m$$

The optimization of SVM

- We can use **Lagrange multipliers** to transform the primal problem it to a dual problem (which means solve an optimization problem by looking at another its related optimization problem).

$$\min \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{s.t.} \quad g_i(\mathbf{w}, b) = 1 - y_i (\mathbf{w}^T \mathbf{x}_i + b) \leq 0, i = 1, 2, \dots, m$$

$$\Downarrow$$

$$L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^m \alpha_i (1 - y_i (\mathbf{w}^T \mathbf{x}_i + b))$$

- For each sample $(\mathbf{x}_i, y_i)$, it has a corresponding Lagrange multiplier $\alpha_i \geq 0$. The $L(\mathbf{w}, b, \boldsymbol{\alpha})$ is called Lagrangian.

(More details in the ESTR3108 lecture note)

The optimization of SVM

- Take the derivative w.r.t. $\mathbf{w}$ and b , and set them equal to zero

$$L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^m \alpha_i (1 - y_i (\mathbf{w}^T \mathbf{x}_i + b)),$$

$$\begin{cases} \frac{\partial L}{\partial \mathbf{w}} = 0 \\ \frac{\partial L}{\partial b} = 0 \end{cases} \implies \begin{cases} \mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i \\ 0 = \sum_{i=1}^m \alpha_i y_i \end{cases}$$

- Then, plug them into $L(\mathbf{w}, b, \boldsymbol{\alpha})$, and construct the dual problem:

$$\max_{\boldsymbol{\alpha}} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

$$\text{s.t. } \sum_{i=1}^m \alpha_i y_i = 0, \alpha_i \geq 0, i = 1, 2, \dots, m$$

The optimization of SVM

- For the dual problem, we also have KKT (Karush-Kuhn-Tucker) ² conditions which are necessary for optimality:

$$\begin{cases} \alpha_i \geq 0 \\ y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 \geq 0 \\ \alpha_i(y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1) = 0 \end{cases}$$

- If $y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 > 0$, it means that the distance between the sample point $(\mathbf{x}_i, y_i)$ is farther than the support vectors, and α_i must be 0.
- If $y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 = 0$, it means that the sample points $(\mathbf{x}_i, y_i)$ is a support vector, and $\alpha_i > 0$.

²https://en.wikipedia.org/wiki/Karush-Kuhn-Tucker_conditions

The optimization of SVM

Dual problem of optimization

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \\ \text{s.t.} \quad & \sum_{i=1}^m \alpha_i y_i = 0, \alpha_i \geq 0, i = 1, 2, \dots, m \end{aligned}$$

- This optimization problem is a **quadratic programming (QP)** problem. There are many effective algorithm can solve this problem, such as **SMO (Sequential Minimal Optimization)**³.

³https://en.wikipedia.org/wiki/Sequential_minimal_optimization

The optimization of SVM

- After obtaining the optimal parameter α^* of the dual problem, recall that we have the derivative w.r.t $\mathbf{w}$ and b , by setting them to zero, we obtain $\mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i$. So that, we can get the optimal $\mathbf{w}^*$ as:

$$\mathbf{w}^* = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i$$

- For the optimal b^* , as we got the $\mathbf{w}^*$ and $y_s (\mathbf{w}^T \mathbf{x}_s + b) = 1$, we can use arbitrary support vector $(\mathbf{x}_s, y_s)$ to get it: $b^* = \frac{1}{y_s} - \mathbf{w}^T \mathbf{x}_s$, where $S = \{\alpha_i > 0, i = 1, 2, \dots, m\}$ are the indices of support vector.
- Practically, we suggest averaging over all eligible support vectors to alleviate numerical error: $b^* = \frac{1}{|S|} \sum_{s \in S} (\frac{1}{y_s} - \mathbf{w}^T \mathbf{x}_s)$.
- Overall, we can get the optimal hyperplane:

$$\begin{cases} \mathbf{w}^* = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i \\ b^* = \frac{1}{|S|} \sum_{s \in S} (\frac{1}{y_s} - \mathbf{w}^T \mathbf{x}_s) \end{cases} \Rightarrow f(x) = \mathbf{w}^T \mathbf{x} + b$$

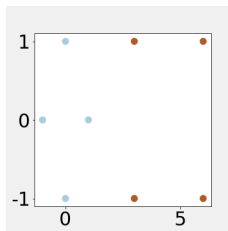
A toy example of SVM

- Assume that we have a set of two-dimensional vector samples, the **positive** data points are:

$$\left\{ \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix} \right\}$$

and the **negative** data points are:

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\}$$



An example of SVM

- Our objective is:

$$\max_{\alpha} \sum_{j=1}^8 \alpha_i - \frac{1}{2} \sum_{i=1}^8 \sum_{j=1}^8 \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \text{ s.t. } \sum_{i=1}^8 \alpha_i y_i = 0, \alpha_i \geq 0$$

$\Downarrow$

$$\begin{aligned} \max_{\alpha} \left[\sum_{j=1}^8 \alpha_i - \frac{1}{2} \alpha_1^2 \cdot 1 \cdot 1 \left((3, 1)^T (3, 1) \right) - \frac{1}{2} \alpha_1 \alpha_2 \cdot 1 \cdot 1 \cdot \right. \\ \left. \left((3, 1)^T \cdot (3, -1) \right) - \dots - \frac{1}{2} \alpha_7 \alpha_8 \cdot (-1) \cdot (-1) \cdot \right. \\ \left. \left((0, -1)^T (-1, 0) \right) - \frac{1}{2} \alpha_8^2 \cdot (-1) \cdot (-1) \left((-1, 0)^T (-1, 0) \right) \right] \\ \text{s.t. } \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 - \alpha_5 - \alpha_6 - \alpha_7 - \alpha_8 = 0, \alpha_i \geq 0 \end{aligned}$$

- Use a SMO toolbox to solve this QP problem and get the optimal:
 $\alpha_1 = \alpha_2 = 0.25, \alpha_5 = 0.5, \alpha_3 = \alpha_4 = \alpha_6 = \alpha_7 = \alpha_8 = 0$
- Then, calculate the $\mathbf{w}^* = \sum_{i=1}^8 \alpha_i y_i \mathbf{x}_i = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

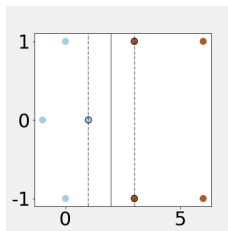
An example of SVM

- There are three support vectors:

$$\mathbf{x}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \mathbf{x}_5 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$$b^* = \frac{1}{|S|} \sum_{s \in S} \left(\frac{1}{y_s} - \mathbf{w}^T \mathbf{x}_s \right) = -2$$

- So we have: $\begin{cases} \mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ b = -2 \end{cases}$
- We can get the hyperplane: $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mathbf{x} - 2 = 0$



Quiz

As we have mentioned in this toy example, we have 8 data points:

$$\left\{ \begin{pmatrix} 7 \\ 1.5 \end{pmatrix}, \begin{pmatrix} 8 \\ -1 \end{pmatrix}, \begin{pmatrix} 6.5 \\ 2 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\}$$

If we have calculated the optimal Lagrange multipliers:

$$\alpha_3 = 0.0266, \alpha_4 = 0.0533, \alpha_5 = 0.0800, \alpha_1 = \alpha_2 = \alpha_6 = \alpha_7 = \alpha_8 = 0$$

Which of the following points are support vectors?

A $\left\{ \begin{pmatrix} 7 \\ 1.5 \end{pmatrix}, \begin{pmatrix} 8 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\}$

B $\left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\}$

C $\left\{ \begin{pmatrix} 6.5 \\ 2 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$

D $\left\{ \begin{pmatrix} 6.5 \\ 2 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix} \right\}$

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D $\left\{ \begin{pmatrix} 6.5 \\ 2 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix} \right\}$

Correct Answer: C

Solution

- As $\alpha_3 = 0.0266, \alpha_4 = 0.0533, \alpha_5 = 0.0800$. These three Lagrange multipliers both be larger than 0.
- For KKT (Karush-Kuhn-Tucker) conditions:

$$\begin{cases} \alpha_i \geq 0 \\ y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 \geq 0 \\ \alpha_i(y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1) = 0 \end{cases}$$

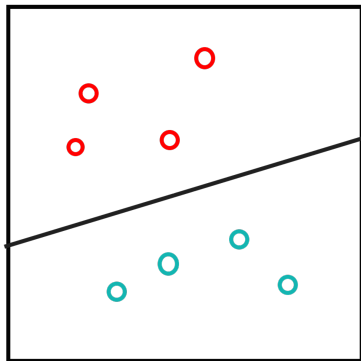
- Recall that if $y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 > 0$, it means that the distance between the sample point $(\mathbf{x}_i, y_i)$ is farther than the support vectors, and α_i must be 0.
- If $y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 = 0$, it means that the sample points $(\mathbf{x}_i, y_i)$ is a support vector and $\alpha_i > 0$.
- So the 3rd, 4th and 5th points are support vectors: $\left\{ \begin{pmatrix} 6.5 \\ 2 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$



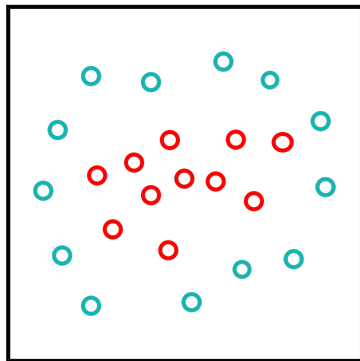
Part 3. Kernel SVM

SVM for non-linear datasets

- Sometimes, we cannot find a straight line to separate the samples.
- In other words, the data is far from being linearly separable in the current dimensions.



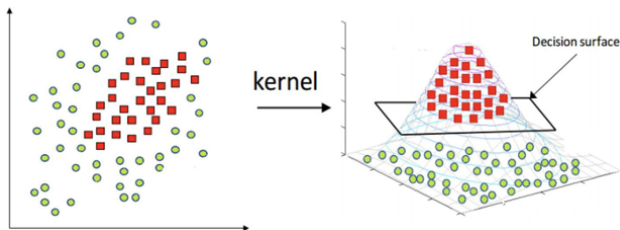
Linearly separable



No good linear separator?

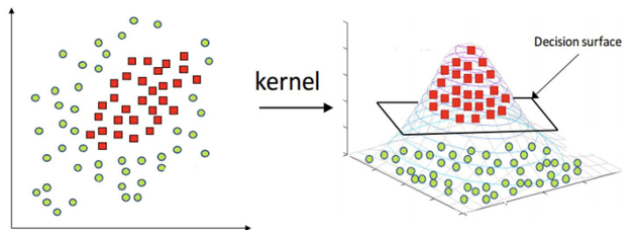
SVM for non-linear datasets

- If we increase the dimension, it may make the inseparable data become separable.
- For example, the samples are in 2D and inseparable, but maybe in 3D, there is a gap between two categories (e.g., a level difference).
- In this case, we can easily draw a separating hyperplane.



SVM for non-linear datasets

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- For example, the samples are in 2D and inseparable, but maybe in 3D, there is a gap between two categories (e.g., a level difference).
- In this case, we can easily draw a separating hyperplane.



- Many ways to increase the feature dimension.

Kernel function in SVM

- SVM can use mathematical functions that are defined as the kernel.
- The function of kernel is to take data as input and transform it into the required form.
- For example, if we have an inseparable two-dimensional data $\begin{pmatrix} x^{(1)} \\ x^{(2)} \end{pmatrix}$,
we can use a transformation: $\phi(\mathbf{x}) = \begin{pmatrix} (x^{(1)})^2 \\ (x^{(2)})^2 \\ \sqrt{2}x^{(1)}x^{(2)} \end{pmatrix}$
- Then we can transform the two-dimensional data to three-dimensional data by $\mathbf{x} \mapsto \phi(\mathbf{x})$, and the data can be linearly separable in a higher-dimensional space:

$$f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b$$

Kernel function in SVM

- The optimization objective for this new feature space should be transformed to:

$$\min \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{s.t.} \quad g_i(\mathbf{w}, b) = 1 - y_i (\mathbf{w}^T \phi(\mathbf{x}_i) + b) \leq 0, i = 1, 2, \dots, m$$

- And the dual problem of optimization should be:

$$\max_{\alpha} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \cancel{\mathbf{x}_i^T \mathbf{x}_j}$$


$$\phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$$

Kernel function in SVM

- However, very large feature spaces have potential issue of memory and computational cost. The “**kernel trick**” can help alleviate this issue.
- It allows us to operate in the original feature space without computing the coordinates of the data in a higher dimensional space.
- Let's assume a function:

$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$$

which means $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ is the **function of $\mathbf{x}_i$ and $\mathbf{x}_j$** , and we don't need to calculate $\phi(\mathbf{x}_i)$ and $\phi(\mathbf{x}_j)$ explicitly. The $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ is called **kernel function**.

Kernel function in SVM

- However, very large feature spaces have potential issue of memory and computational cost. The “**kernel trick**” can help alleviate this issue.
- It allows us to operate in the original feature space without computing the coordinates of the data in a higher dimensional space.
- Let's assume a function:

$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$$

which means $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ is the **function of $\mathbf{x}_i$ and $\mathbf{x}_j$** , and we don't need to calculate $\phi(\mathbf{x}_i)$ and $\phi(\mathbf{x}_j)$ explicitly. The $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ is called **kernel function**.

- For example, quadratic feature mapping for $\mathbf{x} = (x^1, x^2, \dots, x^n) \in \mathbb{R}^n$, $\phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$ has computational $O(n^2)$, while $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ only has computational $O(n)$.

Kernel function in SVM

- Then, the dual problem of optimization can be transformed to:

$$\max_{\alpha} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \cancel{\mathbf{x}_i^T \mathbf{x}_j}$$


$$\cancel{\phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)} \rightarrow \kappa(\mathbf{x}_i, \mathbf{x}_j)$$

- The optimal hyperplane will be:

$$\begin{aligned} f(\mathbf{x}) &= \mathbf{w}^T \phi(\mathbf{x}) + b \\ &= \sum_{i=1}^m \alpha_i y_i \phi(\mathbf{x}_i)^T \phi(\mathbf{x}) + b \\ &= \sum_{i=1}^m \alpha_i y_i \kappa(\mathbf{x}, \mathbf{x}_i) + b \end{aligned}$$

Quiz

Which of the following function $\kappa(\mathbf{x}_i, \mathbf{x}_j)$ can be the kernel function of the transformation:

$$\mathbf{x} = \begin{pmatrix} x^{(1)} \\ x^{(2)} \end{pmatrix} \Rightarrow \phi(\mathbf{x}) = \begin{pmatrix} (x^{(1)})^2 \\ (x^{(2)})^2 \\ \sqrt{2}x^{(1)}x^{(2)} \end{pmatrix}$$

In other words, which one can make $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$?

- A $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j$
- B $\kappa(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i^T \mathbf{x}_j)^2$
- C $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T + \mathbf{x}_j$
- D $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i^T}{\mathbf{x}_j}$
- E $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{2}\mathbf{x}_i^T \mathbf{x}_j$

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- D $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i^T}{\mathbf{x}_j}$
- E $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{2}\mathbf{x}_i^T \mathbf{x}_j$

Correct Answer: B

Quiz solution

- Assume that we have some two-dimensional inseparable data and we use a transformation to transform it to three-dimensional data:

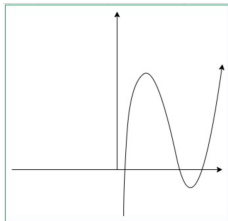
$$\mathbf{x} = \begin{pmatrix} x^{(1)} \\ x^{(2)} \end{pmatrix} \Rightarrow \phi(\mathbf{x}) = \begin{pmatrix} (x^{(1)})^2 \\ (x^{(2)})^2 \\ \sqrt{2}x^{(1)}x^{(2)} \end{pmatrix}$$

- In this case, we can use the a polynomial kernel function:

$$\begin{aligned} \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) &= \begin{pmatrix} (x_i^{(1)})^2 \\ (x_i^{(2)})^2 \\ \sqrt{2}x_i^{(1)}x_i^{(2)} \end{pmatrix}^T \begin{pmatrix} (x_j^{(1)})^2 \\ (x_j^{(2)})^2 \\ \sqrt{2}x_j^{(1)}x_j^{(2)} \end{pmatrix} \\ &= (x_i^{(1)})^2(x_j^{(1)})^2 + (x_i^{(2)})^2(x_j^{(2)})^2 + 2x_i^{(1)}x_j^{(1)}x_i^{(2)}x_j^{(2)} \\ (\mathbf{x}_i^T \mathbf{x}_j)^2 &= \left(x_i^{(1)}x_j^{(1)} + x_i^{(2)}x_j^{(2)} \right)^2 = (x_i^{(1)})^2(x_j^{(1)})^2 + (x_i^{(2)})^2(x_j^{(2)})^2 \\ &\quad + 2x_i^{(1)}x_j^{(1)}x_i^{(2)}x_j^{(2)} = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) \end{aligned}$$

Some popular SVM kernels

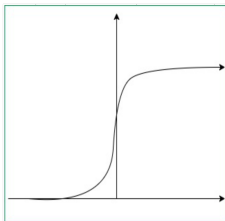
Polynomial Kernel :



$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i^T \mathbf{x}_j)^d$$

It represents the similarity of vectors in training set of data in a feature space over polynomials of the original variables used in kernel.

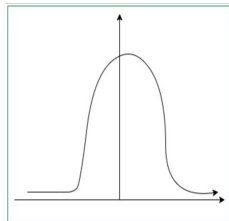
Sigmoid Kernel :



$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \tanh(\beta \mathbf{x}_i^T \mathbf{x}_j + \theta)$$

This function is equivalent to a two-layer, perceptron model of neural network, which is used as activation function for artificial neurons.

Gaussian Kernel :



$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$$

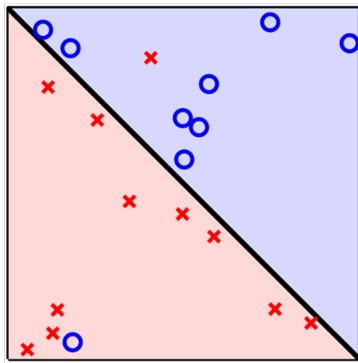
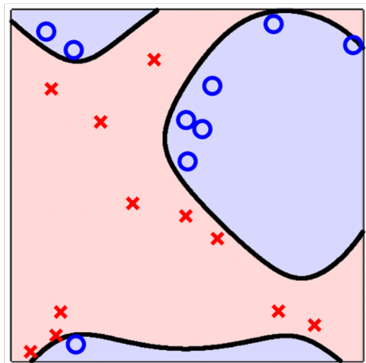
It is used to perform transformation, when there is no prior knowledge about data.



Part 4. Soft-margin SVM

Soft-margin SVM

- Sometimes it is still difficult to find a proper kernel function to make the data **completely separable**.
- Even though we find a kernel function to make the data separable, it may be faced with overfitting problem.
- In this case, we can give up some noisy examples.



Soft-margin SVM

- Recall that the objective is:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

- We could allow some noisy samples not obey the condition

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{s.t.} \quad y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq 1 \text{ for correct } i$$

$$y_i (\mathbf{w}^T \mathbf{x}_i + b) \geq -\infty \text{ for incorrect } i$$

- This idea is called **soft-margin SVM**.

Soft-Margin SVM

- In order to minimize the number of noisy samples, we can use a new **penalty item** and our objective is transformed to:

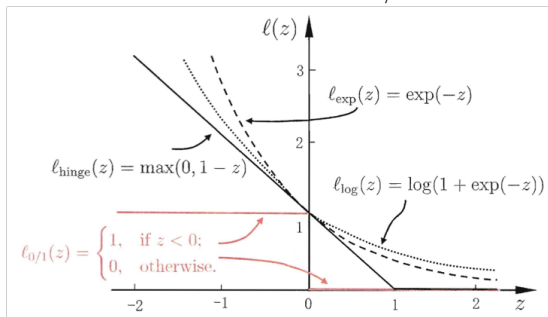
$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^m l_{0/1}(y_i (\mathbf{w}^T \mathbf{x}_i + b) - 1)$$

where C is a constant and $l_{0/1}(z) = 1$ if $z < 0$, otherwise $l_{0/1}(z) = 0$.

- If the sample satisfies the constraint condition, the value of $l_{0/1}$ is 0.
- Otherwise, $l_{0/1}(z) = 1$, and it will be punished.

Soft-margin SVM

- However, the function $l_{0/1}$ is non-convex, discontinuous, which increases the difficulties of optimizing the objective. Therefore, we usually use other functions to replace the $l_{0/1}$ functions:



- For example, if we use hinge loss, the objective can be rewritten to:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^m \max(0, 1 - y_i (\mathbf{w}^T \mathbf{x}_i + b))$$



Part 5. Summary

Summary

- Basic concepts of SVM
- The objective function
- Kernel SVM and Soft-margin SVM

Reading materials

- Slides on **Linear SVM** and **Kernel SVM** by National Taiwan University.
- **Lecture notes** on SVM by Stanford University.
- The first paper proposing SVM, by AT&T Bell Labs, 1995.
<https://link.springer.com/content/pdf/10.1007/bf00994018.pdf>



ESTR3108

Outline

- Supplementary basics of Lagrange multipliers and duality.
- Detailed derivations of optimization for SVM.

Lagrange multipliers

- The method of **Lagrange multipliers** is a powerful technique for solving constrained optimization problems.
- A strategy for finding the extrema (i.e., maxima or minima) of a function subject to some constraints (i.e., the condition that one or more equations have to be satisfied).
- The basic idea is to convert a constrained problem into a form such that the derivative test⁴ of an unconstrained problem is still applied.

⁴https://en.wikipedia.org/wiki/Derivative_test

Lagrange multipliers

- Specifically, consider an optimization problem of the following form:

$$\min_{\boldsymbol{w}} f(\boldsymbol{w})$$

Lagrange multipliers

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$$\min_{\mathbf{w}} \quad f(\mathbf{w})$$

$$\text{s.t.} \quad h_j(\mathbf{w}) = 0, \quad j = 1, \dots, l$$

which means, we want to find the extrema among the set of points $\mathbf{w}$ that satisfy $h_j(\mathbf{w}) = 0, \forall j$.

Lagrange multipliers

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which means, we want to find the extrema among the set of points $\mathbf{w}$ that satisfy $h_j(\mathbf{w}) = 0, \forall j$.

- In the method of Lagrange multipliers, we can define a new objective function (called **Lagrangian**), by introducing a Lagrange multiplier β_i for each constraint, as following:

$$L(\mathbf{w}, \boldsymbol{\beta}) = f(\mathbf{w}) + \sum_{j=1}^l \beta_j h_j(\mathbf{w})$$

- The fact is that extrema of the unconstrained objective L equals to the extrema of the original constrained problem: $L(\mathbf{w}^*, \boldsymbol{\beta}^*) = f(\mathbf{w}^*)$.

Lagrange multipliers

- Now, we will instead find the extrema of L w.r.t. both $\mathbf{w}$ and β .
- Set L 's partial derivatives to zero, and solve for $\mathbf{w}^*$ and β^* :

$$\frac{\partial L}{\partial \mathbf{w}} = \nabla_{\mathbf{w}} f + \sum_{j=1}^l \beta_j \nabla_{\mathbf{w}} h_j = 0$$

$$\frac{\partial L}{\partial \beta_j} = h_j(\mathbf{w}) = 0$$

- In other words, $\mathbf{w}^*$ a local minimum $\Leftrightarrow$ there exists a unique β^* s.t.
 - $\nabla_{\mathbf{w}} L(\mathbf{w}^*, \beta^*) = \mathbf{0}$
 - $\nabla_{\beta} L(\mathbf{w}^*, \beta^*) = \mathbf{0}$ (encodes the equality constraints $h(\mathbf{w}^*) = 0$)
 - $\mathbf{v}^t \nabla_{\mathbf{w}\mathbf{w}}^2 L(\mathbf{w}^*, \beta^*) \mathbf{v} \geq 0 \quad \forall \mathbf{v} \quad \text{s.t.} \quad \nabla_{\mathbf{w}} h(\mathbf{w}^*)^t \mathbf{v} = 0$
(Positive semi-definite Hessian tells us we have a local minimum)

Lagrange multipliers - an example

- Consider the following constrained optimization problem:

$$\begin{array}{ll}\min_{x_1, x_2} & 2x_1^2 + x_2^2 \\ \text{s.t.} & x_1 + x_2 = 1\end{array}$$

- The Lagrangian is:

$$L(x_1, x_2, \beta_1) = 2x_1^2 + x_2^2 + \beta_1(x_1 + x_2 - 1)$$

- Solve for the following partial derivatives:

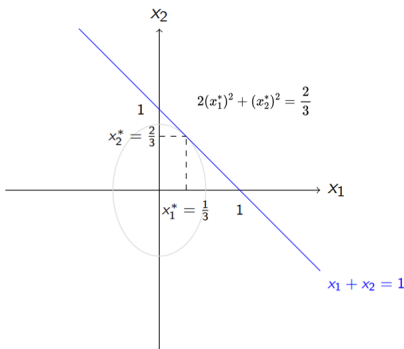
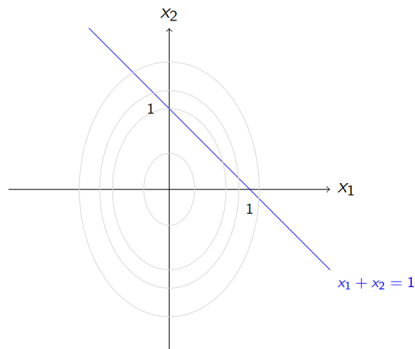
$$\frac{\partial L(x_1^*, x_2^*, \beta_1^*)}{\partial x_1} = 4x_1^* + \beta_1^* = 0$$

$$\frac{\partial L(x_1^*, x_2^*, \beta_1^*)}{\partial x_2} = 2x_2^* + \beta_1^* = 0$$

$$\frac{\partial L(x_1^*, x_2^*, \beta_1^*)}{\partial \beta_1} = x_1^* + x_2^* - 1 = 0$$

Lagrange multipliers - an example

- Then we can obtain: $x_1^* = \frac{1}{3}, x_2^* = \frac{2}{3}, \beta_1^* = -\frac{4}{3}$
- How to understand this optimization problem?



Generalized Lagrangian

- Next, we consider optimization problems that have equality as well as inequality constraints:

$$\begin{array}{ll}\min_{\mathbf{w}} & f(\mathbf{w}) \\ \text{s.t.} & g_i(\mathbf{w}) \leq 0, \quad i = 1, \dots, k + \\ & h_j(\mathbf{w}) = 0, \quad j = 1, \dots, l\end{array}$$

which we'll call the **primal** optimization problem.

- To solve it, we define the generalized Lagrangian as:

$$L(\mathbf{w}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = f(\mathbf{w}) + \sum_{i=1}^k \alpha_i g_i(\mathbf{w}) + \sum_{j=1}^l \beta_j h_j(\mathbf{w}).$$

Here, the α_i and β_j are the Lagrange multipliers.

- Then, let's consider the quantity:

$$\theta_{\mathcal{P}}(\mathbf{w}) = \max_{\alpha \geq 0, \beta} L(\mathbf{w}, \alpha, \beta)$$

Here the $\mathcal{P}$ stands for “primal”.

- Let some $\mathbf{w}$ be given.
 - If it violates any of the primal constraints (e.g., if either $g_i(\mathbf{w}) > 0$ or $h_j(\mathbf{w}) \neq 0$), then:

$$\theta_{\mathcal{P}}(\mathbf{w}) = \max_{\alpha \geq 0, \beta} f(\mathbf{w}) + \sum_{i=1}^k \alpha_i g_i(\mathbf{w}) + \sum_{j=1}^l \beta_j h_j(\mathbf{w}) = \infty$$

- If the constraints are indeed satisfied for a particular value of $\mathbf{w}$, then:

$$\theta_{\mathcal{P}}(\mathbf{w}) = f(\mathbf{w})$$

- Therefore, $\theta_{\mathcal{P}}(\mathbf{w})$ takes the same value as objective in our primal problem for all values of $\mathbf{w}$ that satisfies the primal constraints.

$$\theta_{\mathcal{P}}(\mathbf{w}) = \begin{cases} f(\mathbf{w}) & \text{if } \mathbf{w} \text{ satisfies primal constraints} \\ \infty & \text{otherwise} \end{cases}$$

- Hence, we can write the following minimization problem:

$$\theta_{\mathcal{P}}(\mathbf{w}) = \min_{\mathbf{w}} \max_{\alpha \geq 0, \beta} L(\mathbf{w}, \alpha, \beta)$$

which is the same problem (i.e., has the same solution) as the original, primal problem.

- We define the optimal value of the objective to be $p^* = \min_{\mathbf{w}} \theta_{\mathcal{P}}(\mathbf{w})$, and call this the **value** of the primal problem.

- Next, let's look at another slightly different problem. We define:

$$\theta_{\mathcal{D}}(\alpha, \beta) = \min_{\mathbf{w}} L(\mathbf{w}, \alpha, \beta)$$

Here the $\mathcal{D}$ stands for “dual”.

- We can now pose the **dual** optimization problem:

$$\max_{\alpha \geq 0, \beta} \theta_{\mathcal{D}}(\alpha, \beta) = \max_{\alpha \geq 0, \beta} \min_{\mathbf{w}} L(\mathbf{w}, \alpha, \beta)$$

- This is exactly the same as our primal problem shown above, except that the order of the “max” and the “min” are now exchanged.
- We also define the optimal value of the dual problem's objective as:

$$d^* = \max_{\alpha \geq 0, \beta} \theta_{\mathcal{D}}(\alpha, \beta)$$

Duality

- How are the primal and the dual problems related?
- Consider the intuition, the “max min” of a function always being less than or equal to the “min max”:

$$d^* = \max_{\alpha \geq 0, \beta} \min_{\mathbf{w}} L(\mathbf{w}, \alpha, \beta) \leq \min_{\mathbf{w}} \max_{\alpha \geq 0, \beta} L(\mathbf{w}, \alpha, \beta) = p^*$$

- Under certain conditions, we can have $d^* = p^*$ (called “strong duality”). So that we can solve the dual problem in lieu of the primal problem.

Karush-Kuhn-Tucker (KKT) conditions

- So under which conditions can we solve the dual problem in lieu of the primal problem (i.e., attain $d^* = p^*$) ?
- Suppose that w^* is the solution to the primal problem, α^*, β^* are the solution to the dual problem. The w^*, α^*, β^* must satisfy the **Karush-Kuhn-Tucker (KKT) conditions**⁵:

$$\begin{aligned}\frac{\partial}{\partial w} L(w^*, \alpha^*, \beta^*) &= 0, \\ \frac{\partial}{\partial \beta_j} L(w^*, \alpha^*, \beta^*) &= 0, \quad j = 1, \dots, l \\ \alpha_i^* g_i(w^*) &= 0, \quad i = 1, \dots, k \\ g_i(w^*) &\leq 0, \quad i = 1, \dots, k \\ \alpha_i^* &\geq 0, \quad i = 1, \dots, k\end{aligned}$$

- Then, we have $d^* = p^* = L(w^*, \alpha^*, \beta^*)$.

⁵https://en.wikipedia.org/wiki/Karush-Kuhn-Tucker_conditions

Karush-Kuhn-Tucker (KKT) conditions

- Note the equation $\alpha_i^* g_i(\mathbf{w}^*) = 0$, which is called the KKT dual complementarity condition.
- Specifically, it implies that if $\alpha_i^* > 0$, then $g_i(\mathbf{w}^*) = 0$.
(In other words, the $g_i(\mathbf{w}) \leq 0$ constraint is active, meaning it holds with equality rather than with inequality).
- Later on, this will be used for understanding the support vectors in SVM.

Optimizing SVM

Next, let's go back to details on derivations of SVM.

Optimizing SVM

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- Recall that the optimization problem of SVM is:

$$\min_{\mathbf{w}, b} \quad \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{s.t.} \quad g_i(\mathbf{w}, b) = 1 - y_i (\mathbf{w}^T \mathbf{x}_i + b) \leq 0, i = 1, 2, \dots, m$$

- We have one such constraint for each training sample.

Optimizing SVM

Next, let's go back to details on derivations of SVM.

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- We have one such constraint for each training sample.
- Then, we can construct the Lagrangian:

$$L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^m \alpha_i (1 - y_i (\mathbf{w}^T \mathbf{x}_i + b))$$

- Note that there are only α_i but no β_i Lagrange multipliers, since the problem has only inequality constraints.

Optimizing SVM

- Then, we can transform it to the dual problem:

$$\max_{\alpha} \min_{\mathbf{w}, b} L(\mathbf{w}, b, \alpha) = \max_{\alpha} \min_{\mathbf{w}, b} \left(\frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^m \alpha_i (1 - y_i (\mathbf{w}^T \mathbf{x}_i + b)) \right)$$

- For the Lagrangian, we expand it as:

$$\begin{aligned} L(\mathbf{w}, b, \alpha) &= \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^m \alpha_i (1 - y_i (\mathbf{w}^T \mathbf{x}_i + b)) \\ &= \frac{1}{2} \|\mathbf{w}\|^2 + \sum_{i=1}^m (\alpha_i - \alpha_i y_i \mathbf{w}^T \mathbf{x}_i - \alpha_i y_i b) \\ &= \frac{1}{2} \mathbf{w}^T \mathbf{w} + \sum_{i=1}^m \alpha_i - \sum_{i=1}^m \alpha_i y_i \mathbf{w}^T \mathbf{x}_i - \sum_{i=1}^m \alpha_i y_i b \end{aligned}$$

Optimizing SVM

- To find the dual form of the problem (i.e., to get θ_D), we need to first minimize $\min_{\mathbf{w}, b} L(\mathbf{w}, b, \boldsymbol{\alpha})$ w.r.t. $\mathbf{w}$ and b , for fixed $\boldsymbol{\alpha}$.

$$L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2} \mathbf{w}^T \mathbf{w} + \sum_{i=1}^m \alpha_i - \sum_{i=1}^m \alpha_i y_i \mathbf{w}^T \mathbf{x}_i - \sum_{i=1}^m \alpha_i y_i b$$

- Take the derivative w.r.t. $\mathbf{w}$ and b , and set them equal to zero:

$$\frac{\partial L}{\partial \mathbf{w}} = \frac{1}{2} \times 2 \times \mathbf{w} + \mathbf{0} - \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i - \mathbf{0} = \mathbf{0}$$

$$\Rightarrow \mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i$$

$$\frac{\partial L}{\partial b} = 0 + 0 - 0 - \sum_{i=1}^m \alpha_i y_i = 0$$

$$\Rightarrow \sum_{i=1}^m \alpha_i y_i = 0$$

Optimizing SVM

- As we get $\begin{cases} \mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i \\ 0 = \sum_{i=1}^m \alpha_i y_i \end{cases}$, we plug $\mathbf{w}$ back to the Lagrangian:

$$\begin{aligned} \min_{\mathbf{w}, b} L(\mathbf{w}, b, \boldsymbol{\alpha}) &= \frac{1}{2} \mathbf{w}^T \mathbf{w} + \sum_{i=1}^m \alpha_i - \sum_{i=1}^m \alpha_i y_i \mathbf{w}^T \mathbf{x}_i - \sum_{i=1}^m \alpha_i y_i b \\ &= \frac{1}{2} \mathbf{w}^T \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i - \mathbf{w}^T \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i + \sum_{i=1}^m \alpha_i - b \sum_{i=1}^m \alpha_i y_i \\ &= -\frac{1}{2} \mathbf{w}^T \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i + \sum_{i=1}^m \alpha_i \\ &= -\frac{1}{2} \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i^T \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i + \sum_{i=1}^m \alpha_i \\ &= \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \end{aligned}$$

Optimizing SVM

- Thus, we can get the following dual problem w.r.t. α :

$$\max_{\alpha} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

$$\text{s.t. } \sum_{i=1}^m \alpha_i y_i = 0$$

$$\alpha_i \geq 0, i = 1, \dots, m$$

- Hence, we can solve the dual problem (using Sequential Minimal Optimization, SMO [Platt, 1998]) in lieu of solving the primal problem. After obtaining α^* , we can obtain $\mathbf{w}^*$ and b^* , and get the optimal SVM model $f(\mathbf{x}) = \mathbf{w}^{*T} \mathbf{x} + b^*$.

<https://see.stanford.edu/materials/aimlcs229/cs229-notes3.pdf>

Optimizing SVM

- Verify that the KKT conditions are satisfied:

$$\begin{cases} \frac{\partial L}{\partial \mathbf{w}} = 0 \\ \alpha_i \geq 0 \\ g_i(\mathbf{w}) \leq 0 \\ \alpha_i g(\mathbf{w}) = 0 \end{cases} \Rightarrow \begin{cases} \mathbf{w} = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i \\ \alpha_i \geq 0 \\ y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 \geq 0 \\ \underline{\alpha_i(y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1) = 0} \end{cases}$$

- We can see that if $\alpha_i > 0$, it means $y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 = 0$ and the sample point $(\mathbf{x}_i, y_i)$ is a **support vector**.
- Otherwise, if $\alpha_i = 0$, it means that $y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 > 0$, and the distance is farther than the support vector(s).
- This means, given a sample $\mathbf{x}$, we only need to calculate its inner product with the support vectors (of which there is often only a small number), in order to make predictions with
$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b = \sum_{i=1}^m \alpha_i y_i \mathbf{x}_i^T \mathbf{x} + b$$